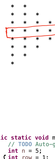
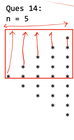
Ques 13:
n = 5



```
public static void main(String[] args) {
    // TODO Auto-generated method stub
    int n = 5;
    int row = 1;
    int star = 1;
    while (row <= 2 * n - 1) {
        // star
        int i = 1;
        while (i <= star) {
            System.out.print("* ");
            i++;
        }

        // mirror

        // next line prep
        System.out.println();
        row++;
    }
}
```

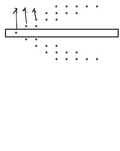
```
public static void main(String[] args) {
    // TODO Auto-generated method stub
    int n = 5;
    int row = 1;
    int star = 1;
    int space = n - 1;
    while (row <= 2 * n - 1) {
        // star
        int i = 1;
        while (i <= star) {
            System.out.print("* ");
            i++;
        }

        // Mirroring
        if (row < n)
            star++;
        else {
            star--;
        }

        // next line prep
        System.out.println();
        row++;
    }
}
```

Ques 14:
n = 5



```
while(row <= 2*n-1) {
    // star
    int i = 1;
    while (i <= star) {
        System.out.print("* ");
        i++;
    }

    // Mirroring
    if (row < n)
        star++;
    else {
        star--;
    }

    // next line prep
    System.out.println();
    row++;
}
```

Ques 16:
n = 5



```
while(row <= 2*n-1) {
    // star
    int i = 1;
    while (i <= star) {
        System.out.print("* ");
        i++;
    }

    // mirror
    if (row <= n-1)
        star--;
    else
        star++;

    // next line prep
    System.out.println();
    row++;
}
```

Ques 19:
n = 7

```
while (row <= 2*n-1) {
    // star
    int i = 1;
    while (i <= star) {
        System.out.print("* ");
        i++;
    }

    // mirror
    if (row <= n-1)
        star--;
    else
        star++;

    // next line prep
    System.out.println();
    row++;
}
```

```
public static void main(String[] args) {
    // TODO Auto-generated method stub
    int n = 5;
    int row = 1;
    int star = 1;
    int space = n / 2;
    int space = n-1;
    while (row <= n) {
        // star
        int i = 1;
        while (i <= star) {
            System.out.print("* ");
            i++;
        }

        // space
        int j = 1;
        while (j <= space) {
            System.out.print(" ");
            j++;
        }

        // mirror
        if (row <= n / 2 + 1) {
            star++;
            space = space - 2;
        } else {
            star++;
            space = space + 2;
        }

        // next line ki prep
        System.out.println();
        row++;
    }

}
}
```

```
public static void main(String[] args) {
    // TODO Auto-generated method stub
    int n = 5;// input
    int row = 1;
    int star = 1;
    int space = n - 1;
    int val=1;
    while (row <= n) {
        // space
        int i = 1;
        while (i <= space) {
            System.out.print("\t");
            i++;
        }

        // star
        int j = 1;
        while (j <= star) {
            System.out.print(val+"\t");
            val++;
            j++;
        }

        // next line ki prep
        star = star + 2;
        space--;
        System.out.println();
        row++;
    }
}
```

Ques 25:
n = 5

```
1
2  3  4
5  6  7  8  9
10 11 12 13 14 15 16
17 18 19 20 21 22 23 24 25
```

```
public static void main(String[] args) {
    // TODO Auto-generated method stub
    int n = 5;// input
    int row = 1;
    int star = 1;
    int space = n - 1;
    int val = 1;
    while (row <= n) {
        // space
        int i = 1;
        while (i <= space) {
            System.out.print(val + "\t");
            val++;
            i++;
        }

        // star
        int j = 1;
        while (j <= star) {
            System.out.print(val+"\t");
            val++;
            j++;
        }

        // next line ki prep
        star = star + 2;
        space--;
        System.out.println();
        row++;
    }
}
```

Ques 27:
n = 5

```
1
1  2  1
1  2  3  2  1
1  2  3  4  3  2  1
```

```
public static void main(String[] args) {
    // TODO Auto-generated method stub
    int n = 5;// input
    int row = 1;
    int star = 1;
    int space = n - 1;
    while (row <= n) {
        // space
        int i = 1;
        while (i <= space) {
            System.out.print("  ");
            i++;
        }

        // star
        int j = 1;
        int val = 1;
        while (j <= star) {
            System.out.print(val + " ");
            if (j < star / 2 + 1) {
                val++;
            } else {
                val--;
            }
            j++;
        }

        // next line ki prep
        star = star + 2;
        space--;
        System.out.println();
        row++;
    }
}
```